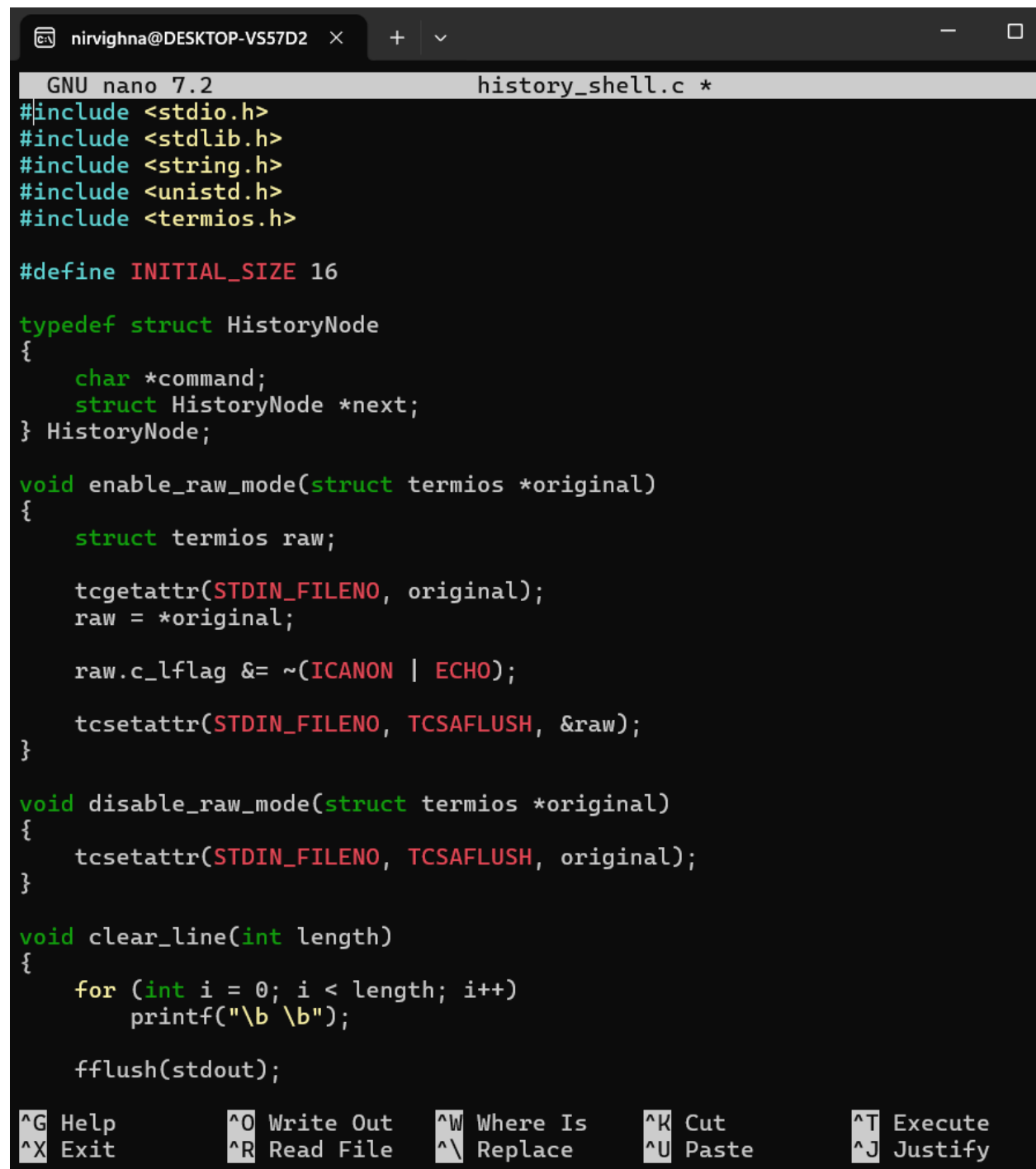


SKILL -3:

```
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ valgrind --version
valgrind-3.22.0
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```



```
GNU nano 7.2 history_shell.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define INITIAL_SIZE 16

typedef struct HistoryNode
{
    char *command;
    struct HistoryNode *next;
} HistoryNode;

void enable_raw_mode(struct termios *original)
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);
    raw = *original;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode(struct termios *original)
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

void clear_line(int length)
{
    for (int i = 0; i < length; i++)
        printf("\b \b");

    fflush(stdout);
}
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^\ Replace	^U Paste	^J Justify

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ ls -l
total 12
-rw-r--r-- 1 nirvighna nirvighna 6065 Sep 21 04:44 history_shell.c
-rw-r--r-- 1 nirvighna nirvighna 1783 Sep 21 04:04 mini_shell.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/history_shell
Interactive History Shell
Type 'exit' to quit.

history-shell> hello
You entered: hello
history-shell> linux
You entered: linux
history-shell> os
You entered: os
history-shell> linux
You entered: linux
history-shell> linux
You entered: linux
history-shell>
```

SKILL -4:

```
nirvighna@DESKTOP-VS57D2  ×  +  ▾
GNU nano 7.2                                parser.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_TOKENS 50
#define MAX_TOKEN_LENGTH 100

typedef struct
{
    char value[MAX_TOKEN_LENGTH];
} Token;

void tokenize(char *input, Token tokens[], int *count)
{
    int i = 0;
    int token_index = 0;

    while (input[i] != '\0' && token_index < MAX_TOKENS)
    {
        /* Skip whitespace */
        while (isspace((unsigned char)input[i]))
        {
            i++;
        }

        if (input[i] == '\0')
            break;

        /* Delimiters */
        if (input[i] == '|' || input[i] == '>' || input[i] == '<')
        {
            tokens[token_index].value[0] = input[i];
            tokens[token_index].value[1] = '\0';

            token_index++;
            i++;
        }
    }
}
```

```
parser.c: Command not found
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ ls -l
total 16
-rw-r--r-- 1 nirvighna nirvighna 6065 Sep 21 04:44 history_shell.c
-rw-r--r-- 1 nirvighna nirvighna 1783 Sep 21 04:04 mini_shell.c
-rw-r--r-- 1 nirvighna nirvighna 3715 Sep 21 05:06 parser.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ cd ~/OS-Week1
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc -Wall -Wextra -g src/parser.c -o bin/parser
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin
total 68
-rwxr-xr-x 1 nirvighna nirvighna 22088 Sep 21 04:45 history_shell
-rwxr-xr-x 1 nirvighna nirvighna 19984 Sep 21 04:05 mini_shell
-rwxr-xr-x 1 nirvighna nirvighna 20992 Sep 21 05:07 parser
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/parser
Simple Command Parser
Type 'exit' to quit.
```

```
parser> ls -l
```

```
--- Token Stream ---
```

```
Token 1: [ls]
```

```
Token 2: [-l]
```

```
Syntax: Valid
```

```
--- Parse Structure ---
```

```
Command 1:
```

```
ls
```

```
-l
```

```
parser> |
```

Syntax: Valid

--- Parse Structure ---

Command 1:

ls

Command 2:

grep

txt

parser> cat < input.txt

--- Token Stream ---

Token 1: [cat]

Token 2: [<]

Token 3: [input.txt]

Syntax: Valid

--- Parse Structure ---

Command 1:

cat

<

input.txt

parser> ls > output.txt

--- Token Stream ---

Token 1: [ls]

Token 2: [>]

Token 3: [output.txt]

Syntax: Valid

--- Parse Structure ---

Command 1:

ls

>

output.txt

parser> |

```
parser> | ls
```

```
--- Token Stream ---
```

```
Token 1: [ ]
```

```
Token 2: [ls]
```

```
Error: Command cannot start with '|'.  
Syntax: Invalid
```

```
parser> ls |
```

```
--- Token Stream ---
```

```
Token 1: [ls]
```

```
Token 2: [ ]
```

```
Error: Command cannot end with '|'.  
Syntax: Invalid
```

```
parser> ls || grep
```

```
--- Token Stream ---
```

```
Token 1: [ls]
```

```
Token 2: [ ]
```

```
Token 3: [ ]
```

```
Token 4: [grep]
```

```
Error: Consecutive pipes are not allowed.  
Syntax: Invalid
```

```
parser> ^[[2~|
```

```
parser>
Empty command.
parser> ls -l | grep txt

--- Token Stream ---
Token 1: [ls]
Token 2: [-l]
Token 3: [|]
Token 4: [grep]
Token 5: [txt]

Syntax: Valid

--- Parse Structure ---
Command 1:
  ls
  -l
Command 2:
  grep
  txt

parser> |
```

```
nirvighna@DESKTOP-VS57D2  x  +  v  -  □  ×
GNU nano 7.2  Makefile *
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGETS = bin/process_demo bin/mini_shell bin/history_shell bin/parser

all: $(TARGETS)

bin/process_demo: src/process_demo.c
    $(CC) $(CFLAGS) src/process_demo.c -o bin/process_demo

bin/mini_shell: src/mini_shell.c
    $(CC) $(CFLAGS) src/mini_shell.c -o bin/mini_shell

bin/history_shell: src/history_shell.c
    $(CC) $(CFLAGS) src/history_shell.c -o bin/history_shell

bin/parser: src/parser.c
    $(CC) $(CFLAGS) src/parser.c -o bin/parser

clean:
    rm -f $(TARGETS)
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ make
make: *** No rule to make target 'src/process_demo.c', needed by 'bin/proces
s_demo'. Stop.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l
total 16
-rw-r--r-- 1 nirvighna nirvighna 498 Sep 21 05:14 Makefile
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 21 05:15 bin
-rw-r--r-- 1 nirvighna nirvighna 1940 Sep 21 04:00 mini_shell.c
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 21 05:06 src
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```